DPA

Density Peak Advanced Clustering

d'Errico, Facco, Laio & Rodriguez (2021)

Information Sciences

Unsupervised Learning Course

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Outline

1. Motivation & Problem
2. **Understanding the Method**
    - TWO-NN (Intrinsic Dimension)
    - PAk (Adaptive Density Estimation)
    - Heuristics 1-3
3. **Hyperparameters & Limitations**
4. **Comparison with Other Methods**
5. Results & Demo

The Problem

Standard Density Peaks (DP) Limitations

DP (Rodriguez & Laio, 2014):

- Manual center selection
- Fixed k-NN density
- No error quantification
- Manual cluster merging

DPA (2021) solves all:

- Automatic center detection
- Adaptive PAk density
- Fisher Information bounds
- Statistical merging (Z-score)

DPA Algorithm Overview

1. **Estimate intrinsic dimension** d using TWO-NN
2. **Compute adaptive density** ρ using PAk
3. **Find cluster centers** (Heuristic 1)
4. **Find saddle points** (Heuristic 2)
5. **Merge insignificant peaks** (Heuristic 3)
6. **Assign points** following density gradient

Key advantage: Only ONE parameter (Z)!

Understanding the Method

1. TWO-NN: Intrinsic Dimension

Estimates the **intrinsic dimension** d of the data manifold.

Idea: For uniform density, the ratio $\mu = r_2/r_1$ follows:

$$P(\mu) = d \cdot \mu^{-d-1}$$

where r_1, r_2 are distances to 1st and 2nd nearest neighbors.

Assumes:

- Locally constant density up to the 2nd neighbor
- i.i.d. points

Understanding the Method

2. PAk: The Statistical Model

Shell volumes:

$$v_{i,l} = \omega_d \, (r_{i,l}{}^d - r_{i,l-1}{}^d)$$

ω_d = volume of the unit d-ball; $r_{i,l}$ = ordered distances to the neighbors of i; d = intrinsic dimension from TWO-NN

Key observation: if the density is constant, the $v_{i,l}$ are i.i.d. exponential with rate ρ.

Log-likelihood:

$$L_{i,k}(\rho_i) = k \cdot \log \rho_i - \rho_i \cdot \Sigma_{l=1}{}^k \, v_{i,l} = k \cdot \log \rho_i - \rho_i \cdot V_{i,k}$$

maximization gives the k-NN estimator; variance decreases as $1/\sqrt{k}$

Understanding the Method

2. PAk: Optimal $\hat{k}$ and Error

Likelihood Ratio Test between two models:

- (1) densities at the point and at its k-th neighbor are the same
- (2) the densities are assumed to be different

Stopping criterion:

$$\hat{k}_i : (D_k < D_{thr} \ \forall\ k \leq \hat{k}_i) \ \& \ (D_{\hat{k}_i+1} \geq D_{thr})$$

D_{thr} = 23.928, corresponding to a p-value of 10^{-6}

Systematic correction: log(ρ) replaced by log(ρ) + α·l — α models the linear density trend

Error from Fisher Information:

$$\varepsilon_i = \sqrt{4(\hat{k}_i+2) / ((\hat{k}_i-1)\cdot\hat{k}_i)}$$

Understanding the Method

3. The Three Heuristics

Heuristic	Purpose	Criterion
H1	Find cluster centers	1. $\delta_i > r_{\hat{k}_i}$ 2. $\forall\, j,\, g_j > g_i : i \notin NN_j := \{k : r_{jk} < r_{\hat{k}_j}\}$
H2	Find saddle points	1. $j = \text{argmin}_{k \in c'}\, r_{ik}$, with $r_{ij} < r_{\hat{k}_i}$ 2. $i = \text{argmin}_{k \in c}\, r_{jk}$
H3	Merge peaks	Z-test on peak vs saddle

$g = \log(\rho) - \varepsilon$ (error-adjusted density)

Hyperparameters

The Z Parameter

Merge criterion (Heuristic 3):

$$\log(\rho_c) - \log(\rho_{cc'}) < Z \cdot (\varepsilon_c + \varepsilon_{cc'})$$

ρ_c = density of the center of cluster c; $\rho_{cc'}$ = density of the saddle between c and c'

Z	Effect
1.0	More clusters (liberal)
2.0	Balanced (default)
3.0	Fewer clusters (conservative)

Limitations

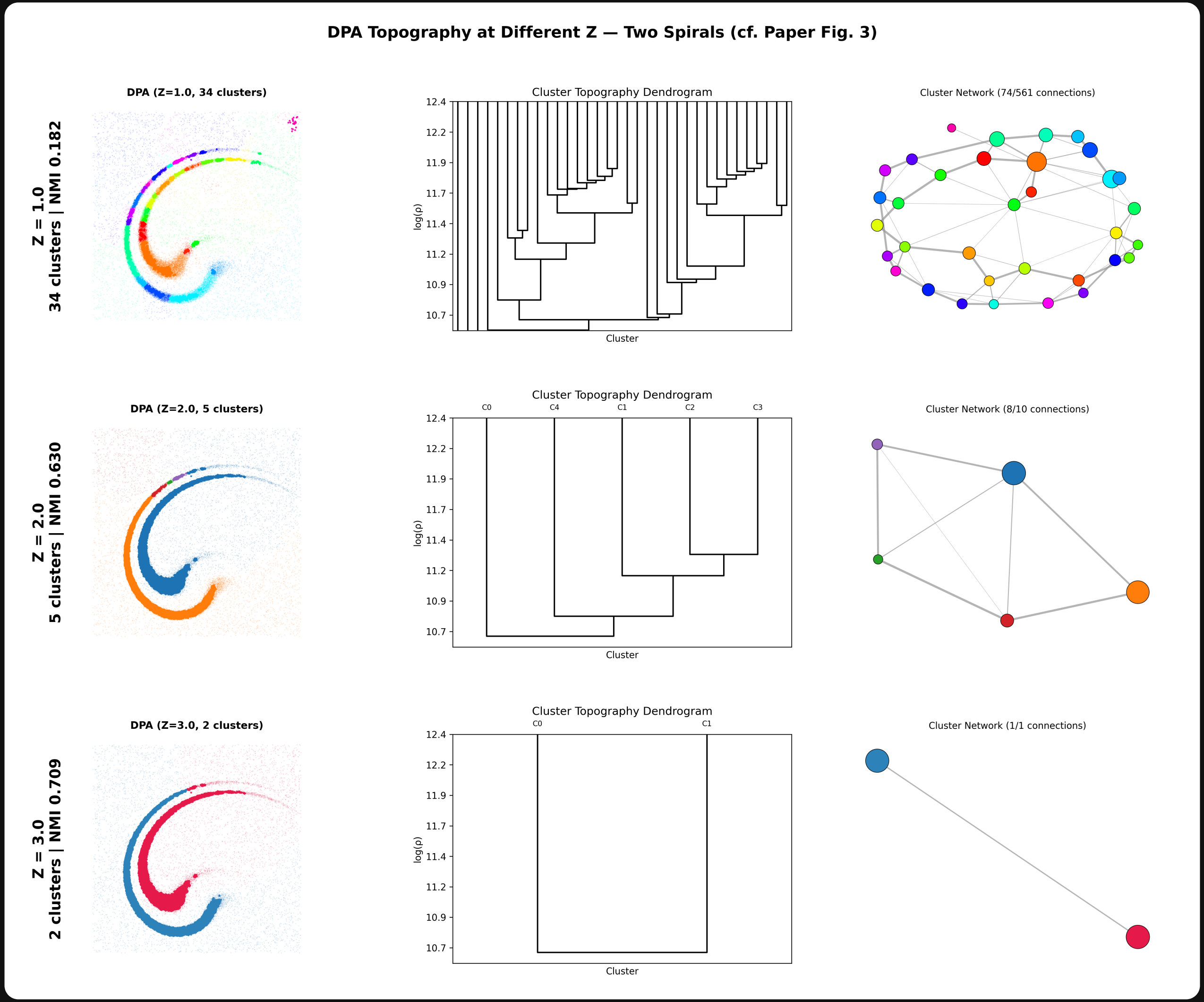
Weaknesses

- **High dimensions (d > 15):** PAk estimation degrades
- **Computation:** $O(N \times k_{max} \times \log N)$
- **Uniform density:** No natural peaks

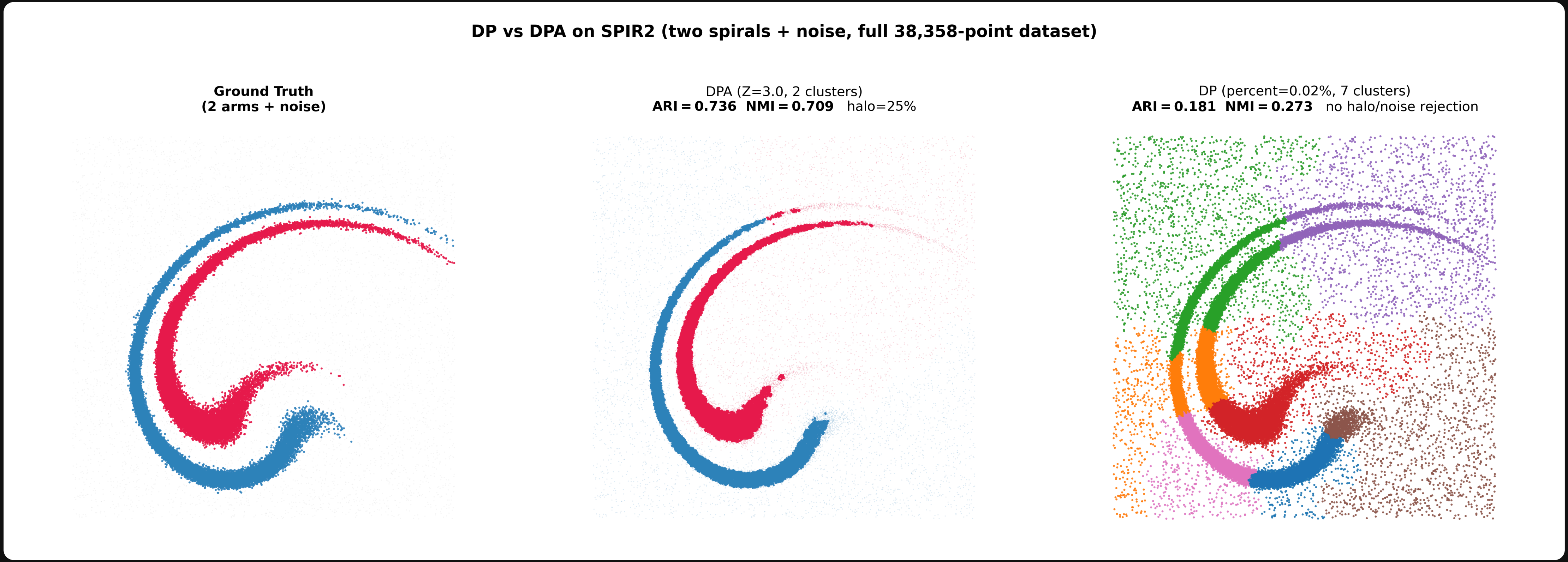
Mitigations

- Use lower Z for high-D
- FAISS for fast k-NN
- Check decision graph

SPIR2 - DPA topography



DPA vs DP


DP vs DPA on SPIR2 (two spirals + noise, full 38,358-point dataset)
Ground Truth
(2 arms + noise)
DPA (Z=3.0, 2 clusters)
ARI = 0.736 NMI = 0.709 halo=25%
DP (percent=0.02%, 7 clusters)
ARI = 0.181 NMI = 0.273 no halo/noise rejection

Conclusions

- **DPA automates** what DP required manually
- **Single parameter Z** with clear statistical meaning
- **Error quantification** via Fisher Information
- **Topographic analysis** for cluster interpretation

Thank You!

Questions?

Code: github.com/francescacraievich/Data-Topography-DPA